

API Specification Doc

“Fábula Dental” Management System

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Index

Content

1. Register a new patient.....	5
Registers a new patient with ID, name, birthday, and reason for consultation.....	5
Request.....	5
Response.....	5
2. Get all patients.....	6
Request.....	6
Response.....	6
3. Update patient information.....	6
Updates an existing patient's information directly from the table.....	6
Request.....	6
Response.....	6
4. Delete a patient.....	7
Deletes a patient record.....	7
Request.....	7
Response.....	7
5. Render patient registration form.....	8
Renders the visual form to register a new patient.....	8
Request.....	8
Response.....	8
6. Render patient edit form.....	8
Loads the edit form pre-filled with the patient's existing data.....	8
Request.....	8
Response.....	8
7. Record a payment.....	9
Records a new patient payment (Deposit or Final).....	9
Request.....	9
Response.....	9
8. Get payment history.....	9
Retrieves the payment history and calculates current status (Pending, Partial, Completed).....	9
Request.....	9
Response.....	10
9. Get payment history.....	10
Updates a recorded payment to correct errors or update status.....	10
Request.....	10
Response.....	10
10. Register a dental supply.....	11
Registers a new dental supply into the inventory with quantity and dates.....	11

Request.....	11
Response.....	11
11. Get inventory list.....	11
Retrieves the inventory list, automatically calculating expiration status (Valid, Due, Expired).....	11
Request.....	11
Response.....	12
12. Update supply quantity.....	12
Edits an existing supply in the inventory (e.g., updating quantity).....	12
Request.....	12
Response.....	12
13. Render payment form.....	13
Renders the visual form to record payments.....	13
Request.....	13
Response.....	13
14. Render supply management form.....	13
Renders the visual form to manage inventory.....	13
Request.....	13
Response.....	13
15. Render public treatments page.....	13
Public page showing available treatments, accessible without login.....	13
Request.....	14
Response.....	14
16. Render login page.....	14
Main entry point / Login authentication for roles.....	14
Request.....	14
Response.....	14
17. Delete a payment record.....	14
Deletes a payment record from history.....	14
Request.....	14
Response.....	15
18. Delete a dental supply.....	15
Removes a dental supply from the inventory.....	15
Request.....	15
Response.....	15
19. Render payment edit form.....	16
Loads the edit form pre-filled with an existing payment record.....	16
Request.....	16
Response.....	16
20. Render supply edit form.....	16
Loads the edit form pre-filled with an existing supply item.....	16
Request.....	16

Response.....	17
21. Render success page.....	17
Displays a success confirmation page after a create or update operation.....	17
Request.....	17
Response.....	17
22. Render error page.....	18
Displays an error page when a validation or database operation fails.....	18
Response.....	18
23. Render administrator dashboard.....	18
Renders the administrator dashboard with access to all management modules.....	18
Request.....	18
Response.....	19
24. Render dentist dashboard.....	19
Response.....	19
25. Render receptionist dashboard.....	19
Response.....	19
Conventions.....	21
Status Codes.....	21

Methods

1. Register a new patient

Registers a new patient with ID, name, birthday, and reason for consultation.

Request

Method	URL
POST	/api/patients

Type	Params	Values
POST	patientID	string
POST	fullName	string
POST	birthday	string
POST	reasonForConsultation	string

Response

Status	Response
201	{"success": true, "message": "Patient registered successfully"}
400	{"error": "Missing required fields."}
500	{"error": "Something went wrong. Please try again later."}
201	{"success": true, "message": "Patient registered successfully"}

2. Get all patients

Retrieves the list of all registered patients to populate the dentist's table dynamically.

Request

Method	URL
GET	/api/patients

Response

Status	Response
200	[{"patientID": "...", "fullName": "...", "phone": "..."}]
400	{"error": "Unable to fetch patients."}

3. Update patient information

Updates an existing patient's information directly from the table.

Request

Method	URL
PUT	/api/patients/:patientId

Type	Params	Values
PUT	patientID	string
PUT	phone	string
PUT	reasonForConsultation	string

Response

Status	Response
200	<code>{"success": true, "message": "Patient updated"}</code>
400	<code>{"error": "Patient ID is required."}</code>
404	<code>{"error": "Patient not found."}</code>
500	<code>{"error": "Something went wrong."}</code>

4. Delete a patient

Deletes a patient record.

Request

Method	URL
DELETE	<code>/api/patients/:patientId</code>

Type	Params	Values
URL_PARAM	id	number

Response

Status	Response
200	<code>{"success": true, "message": "Patient deleted"}</code>
400	<code>{"error": "Invalid or missing patient ID."}</code>
404	<code>{"error": "Patient not found."}</code>
500	<code>{"error": "Deletion failed."}</code>

5. Render patient registration form

Renders the visual form to register a new patient.

Request

Method	URL
GET	/patients/new

Response

Status	Response
200	Returns HTML view (patient registration form)
500	{"error": "Unable to load form."}

6. Render patient edit form

Loads the edit form pre-filled with the patient's existing data.

Request

Method	URL
GET	/patients/:patientId/edit

Type	Params	Values
URL_PARAM	patientID	string

Response

Status	Response
200	Returns HTML view (pre-filled patient edit form)
400	{"error": "Patient ID required."}

404	{"error": "Patient not found."}
500	{"error": "Unable to load edit form."}

7. Record a payment

Records a new patient payment (Deposit or Final).

Request

Method	URL
POST	/api/payments

Type	Params	Values
POST	patientID	string
POST	amount	number
POST	paymentType	["Deposit" "Final"]
POST	paymentMethod	string

Response

Status	Response
201	{"success": true, "message": "Payment recorded"}
400	{"error": "Invalid payment data."}
500	{"error": "Could not record payment."}

8. Get payment history

Retrieves the payment history and calculates current status (Pending, Partial, Completed).

Request

Method	URL
GET	/api/payments

Response

Status	Response
200	[{"paymentID": "...", "amount": 50.0, "status": "Partial"}]
500	{"error": "Unable to fetch payment history."}

9. Get payment history

Updates a recorded payment to correct errors or update status.

Request

Method	URL
PUT	/api/payments/:paymentId

Type	Params	Values
PUT	paymentID	string
PUT	amount	number
PUT	status	["Pending" "Partial" "Completed"]

Response

Status	Response
200	{"success": true, "message": "Payment updated"}
400	{"error": "Invalid payment ID or data."}
404	{"error": "Payment record not found."}

10. Register a dental supply

Registers a new dental supply into the inventory with quantity and dates.

Request

Method	URL
POST	/api/supplies

Type	Params	Values
POST	supplyName	string
POST	quantity	number
POST	orderDate	string
POST	expirationDate	string

Response

Status	Response
201	{"success": true, "message": "Supply added"}
400	{"error": "Missing supply data."}
500	{"error": "Could not add supply."}

11. Get inventory list

Retrieves the inventory list, automatically calculating expiration status (Valid, Due, Expired).

Request

Method	URL
GET	/api/supplies

Response

Status	Response
200	[{"supplyID": "...", "supplyName": "Resin", "status": "Current"}]
500	{"error": "Unable to fetch inventory."}

12. Update supply quantity

Edits an existing supply in the inventory (e.g., updating quantity).

Request

Method	URL
PUT	/api/supplies/:supplyId

Type	Params	Values
PUT	supplyID	string
PUT	quantity	number

Response

Status	Response
200	{"success": true, "message": "Supply updated"}
400	{"error": "Invalid supply ID or quantity."}
404	{"error": "Supply not found."}

13. Render payment form

Renders the visual form to record payments.

Request

Method	URL
GET	/payments/new

Response

Status	Response
200	Returns HTML view (payment form)
500	{"error": "Unable to load form."}

14. Render supply management form

Renders the visual form to manage inventory.

Request

Method	URL
GET	/supplies/new

Response

Status	Response
200	Returns HTML view (supply management form)
500	{"error": "Unable to load form."}

15. Render public treatments page

Public page showing available treatments, accessible without login.

Request

Method	URL
GET	/treatments

Response

Status	Response
200	Returns HTML view (public treatments page)
500	{"error": "Unable to load page."}

16. Render login page

Main entry point / Login authentication for roles.

Request

Method	URL
GET	/login

Response

Status	Response
200	Returns HTML view (login page)
500	{"error": "Unable to load login page."}

17. Delete a payment record

Deletes a payment record from history.

Request

Method	URL
--------	-----

DELETE	<code>/api/payments</code>
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Type	Params	Values
DELETE	id	string

Response

Status	Response
200	<code>{"success": true, "message": "Payment deleted"}</code>
400	<code>{"error": "Payment ID required."}</code>
404	<code>{"error": "Payment not found."}</code>
500	<code>{"error": "Deletion failed."}</code>

18. Delete a dental supply

Removes a dental supply from the inventory.

Request

Method	URL
DELETE	<code>/api/supplies/:supplyId</code>

Type	Params	Values
DELETE	id	string

Response

Status	Response
200	<code>{"success": true, "message": "Supply deleted"}</code>
400	<code>{"error": "Supply ID required."}</code>

404	{"error": "Supply not found."}
500	{"error": "Deletion failed."}

19. Render payment edit form

Loads the edit form pre-filled with an existing payment record.

Request

Method	URL
GET	/payments/:paymentId/edit

Type	Params	Values
URL_PARAM	id	string

Response

Status	Response
200	Returns HTML view (pre-filled Payment edit form)
400	{"error": "Payment ID required."}
404	{"error": "Payment not found."}
500	{"error": "Unable to load edit form."}

20. Render supply edit form

Loads the edit form pre-filled with an existing supply item.

Request

Method	URL
--------	-----

GET	/supplies/:supplyId/edit
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Type	Params	Values
URL_PARAM	id	string

Response

Status	Response
200	Returns HTML view (pre-filled supply edit form)
400	{"error": "Supply ID required."}
404	{"error": "Supply not found."}
500	{"error": "Unable to load edit form."}

21. Render success page

Displays a success confirmation page after a create or update operation.

Request

Method	URL
GET	/:entity/success

Type	Params	Values
URL_PARAM	type	["patient" "payment" "supply"]

Response

Status	Response
200	Returns HTML view (success confirmation page)
400	{"error": "Invalid entity type."}

500	{"error": "Unable to load success page."}
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22. Render error page

Displays an error page when a validation or database operation fails.

Request

Method	URL
GET	/error

Type	Params	Values
URL_PARAMETER	type	["patient" "payment" "supply"]

Response

Status	Response
200	Returns HTML view (error page)
400	{"error": "Invalid entity type."}
500	{"error": "Unable to load error page."}

23. Render administrator dashboard

Renders the administrator dashboard with access to all management modules.

Request

Method	URL
GET	/admin/dashboard

Response

Status	Response
200	Returns HTML view (administrator dashboard)
401	{"error": "Unauthorized access."}
500	{"error": "Unable to load dashboard."}

24. Render dentist dashboard

Renders the dentist dashboard with access to patient management.

Request

Method	URL
GET	/dentist/dashboard

Response

Status	Response
200	Returns HTML view (dentist dashboard)
401	{"error": "Unauthorized access."}
500	{"error": "Unable to load dashboard."}

25. Render receptionist dashboard

Renders the receptionist dashboard with access to payments and scheduling.

Request

Method	URL
GET	/receptionist/dashboard

Response

Status	Response
200	Returns HTML view (receptionist dashboard)
401	{"error": "Unauthorized access."}
500	{"error": "Unable to load dashboard."}

Glossary

Conventions

- **Client** - Client application.
- **Status** - HTTP status code of response.
- All responses are in JSON format except where HTML views are returned.
- All request parameters are mandatory unless marked as [optional].
- All request parameters are mandatory unless explicitly marked as [optional]
- The API includes both RESTful endpoints (data operations) and HTML views (forms and dashboards).
- Authentication is required for most endpoints except the public treatments page and login page.

Status Codes

All status codes are standard HTTP status codes. The below ones are used in this API.

2XX - Success of some kind

4XX - Error occurred in client's part

5XX - Error occurred in server's part

Status Code	Description
200	OK
201	Created
400	Bad request
401	Authentication failure
403	Forbidden
404	Resource not found
500	Internal Server Error